

Cognitive Testing During Mild Acute Ischemic Stroke Predicts Long-Term Return to Work

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Objectives: Many survivors of a mild ischemic stroke do not return to work or driving. Cognitive testing is commonly done to assess long-term cognitive impairment after stroke. Inpatient cognitive testing during the acute period of ischemic stroke may also be a predictor for workforce reengagement and functional outcome. **Materials and methods:** At our comprehensive stroke center, we prospectively enrolled previously working adults < 65 years old who were diagnosed with first-ever ischemic stroke, had a prestroke modified Rankin Scale (mRS) ≤ 1 and NIHSS ≤ 3 . Testing performed within 1 week of stroke included the Montreal Cognitive Assessment (MOCA), Clock Drawing Test (CDT), Trail Making Tests A and B, Backward Digit Span Test, and Hospital Anxiety and Depression Scale (HADS). Other data obtained included age, gender, years of education, occupation, stroke location, stroke laterality, and presence of white matter disease on imaging. Outcome measures assessed at 3 months, 6 months, and 12 months post-stroke included return to work, return to driving, and mRS. In a logistic regression analysis, we performed both univariate and multivariate analyses. Multivariate analysis was completed on variables with $p\text{-value} \leq 0.05$ in the univariate analysis. **Results:** Of 39 total stroke patients enrolled and tested (median [IQR] age 55 [46–60] years; 77.5% male; 22.5% female), 36 completed 3-month follow up, of which 58% returned to work, 78% returned to driving, and 72% had mRS of 0–1. In multivariate analysis, a single point increase in the clock drawing task score increased the odds of return to work by 3.79 (95% CI, 1.10–14.14) and return to driving by 6.74 (95% CI, 1.22–37.23) at 3 months. MOCA and HADS were both associated with mRS ≤ 1 . MOCA was associated with return to work at 6 months and CDT was associated with return to work at 12 months. **Conclusion:** Cognitive testing with CDT and MOCA in the acute period after ischemic stroke may predict common patient goals post stroke, including return to work, driving, and independence. These tools can potentially be used for prognosis and identifying those who may benefit from further interventions.

Key Words: Acute ischemic stroke—Mild stroke—Cognitive testing—Outcomes
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Introduction

Ischemic stroke is a leading cause of disability worldwide, often affecting working-aged adults and requiring intensive rehabilitation.^{1–3} Population based studies show that most cases of stroke are mild with a National Institutes of Health Stroke Scale (NIHSS) of less than 6, with a median score of 3 in one study.^{4,5} Even in cases of mild stroke, there is often cognitive dysfunction that may interfere with working and recovery of full function.^{6,7} A study of mild to moderate ischemic stroke with NIHSS 0–16 found that over a third never returned to work, and of those that did, over half were not able to perform at full function.⁸ There is also impairment of driving ability after stroke, even in mild stroke.^{7,9}

Finding predictors of return to work and driving immediately after a mild ischemic stroke would benefit patient and family planning as well as directing rehabilitation services. There is evidence that neuropsychological evaluation during acute stroke can predict functioning at work and social domains 12 months post-stroke.¹⁰ However, full neuropsychological evaluations and task performance assessments are lengthy and often not possible to administer immediately post-stroke in the inpatient setting. The results of more feasible bedside cognitive testing during acute ischemic stroke, including the Montreal Cognitive Assessment (MOCA) and the Clock Drawing Test (CDT), has been associated with long-term cognitive impairment, functional independence, and degree of reintegration into daily occupations up to 1-year post-stroke in previous studies.^{11–13} In this exploratory study, we aim to assess whether cognitive testing during the acute period of mild ischemic stroke can predict return to work, driving, and independent function over a year.

Methods

From July 2018 to June 2019, ischemic stroke patients were prospectively recruited from our comprehensive stroke center's inpatient stroke service, located in a major metropolitan area in the midwestern United States. This study was approved by our organization's Institutional Review Board. Diagnosis of stroke was determined by imaging criteria or in the absence of imaging-confirmed diagnosis, by clinical diagnosis by the primary vascular neurologist. Mild stroke was defined as NIHSS of 3 or less at time of enrollment, as defined by previous studies.^{13,14} Patients were eligible if they fulfilled the following criteria: age 18–65, prestroke modified Rankin Scale (mRS) 0–1, first-ever ischemic stroke, work experience at least 1 year immediately prior to stroke, and enrollment within 1 week of stroke. Exclusion criteria were predicted life expectancy of less than 1 year, not planning or desiring to return to work, or completely resolved deficits from stroke prior to testing.

Eligible patients were consented while hospitalized. Data collected included age, gender, years of education,

occupation and occupation type (blue-collar or white-collar), NIHSS, prestroke mRS, stroke location (cortical, subcortical, or posterior), stroke laterality, and presence of white matter disease on imaging. Examples of blue-collar occupation types among participants included “construction worker” and “house cleaner”, and examples of white-collar occupation types included “real estate broker” and “lawyer”. The NIHSS is a 15-item impairment scale used to measure stroke severity, which is a widely used outcomes measure in the recombinant tissue plasminogen activator stroke trials.¹⁵ The mRS is a marker for disability ranging from 0–6, which is similarly widely used in stroke outcome studies.¹⁶ For statistical analysis, NIHSS was treated as a continuous variable and mRS was treated as an ordinal variable. The NIHSS, prestroke mRS, and stroke location and laterality were determined by the vascular neurologist treating the patient. The presence of white matter disease was determined by the reading radiologist.

Bedside cognitive testing using the following tools was performed by one of two study coordinators trained by a clinical neuropsychologist. The Montreal Cognitive Assessment (MOCA) is a cognitive screening tool, which is scored out of 30 points and tests a broad range of cognitive domains, is devised to differentiate between normal aging and mild cognitive impairment, and has been widely used clinically in a broad range of neurologic diseases.¹⁷ For statistical analysis, MOCA was treated as a continuous variable, as established cutoff scores on MOCA have varied in previous literature, with difficulty establishing optimal sensitivities and specificities for cutoffs.^{18,19} The MOCA includes small sections on clock drawing and trail making, however these visuospatial tasks were also measured in stand-alone tests to expand sensitivity for this domain. The CDT requires participants to copy and then produce a clock, and was scored from 0–10, which is a method developed by Shulman²⁰ and validated in previous literature.^{21,22} The CDT was analyzed as a continuous variable. The Trail Making Test A (TMT-A) consists of 25 circles (numbered 1–25) distributed over a piece of paper, with instructions to connect the numbers in ascending order as quickly as possible and scored based on time to complete. The Trail Making Test B (TMT-B) also includes 25 circles distributed over a piece of paper but includes both numbers^{1–13} and letters (A–L). The participant is again instructed to, as quickly as possible, connect the circles in an ascending order, but with the added task of alternating between the numbers and letters (*i.e.*, 1-A-2-B-3-C, etc.).²³ The times to complete TMT-A and TMT-B were treated as continuous variables. The backward digit spanning test was devised for this study as two sets each of an increasing sequence of digits from two digits to five digits, which participants were instructed to recall and recite backwards. This is a common tool to assess working memory in age and neuropsychological disease, and it was treated as a continuous

variable.²⁴ The Hospital and Anxiety Depression Scale (HADS) is a questionnaire and has been found to be a reliable measure for detection of depression and anxiety in the setting of nonpsychiatric medical diagnoses.²⁵ This was scored as either normal or abnormal with cutoff score of 8 or higher out of 21 points, and it was treated as a categorical variable.

Outcome measures were assessed at +/- 15 days at the following time points of 3, 6 and 12 months post stroke via telephone by investigator, who was blinded to the subject's score on any element of cognitive testing done during hospitalization to protect against bias. Outcome measures included return to work, return to driving, and functional independence with an mRS ≤ 1 , and were analyzed as categorical variables.

Data analysis

The target subject enrollment was 40, determined based on power analysis to detect a 3-point difference in the MOCA. Participants' general and clinical characteristics were described using frequency (percentage) count. For further analysis, the appropriate statistics was determined based on the outcome normality of testing using Kolomogorov-Smirnov test. Univariate analysis was done by comparing each predictor variable with each of the outcome measures. Only significant variables ($p \leq 0.05$) from univariate analyses were included in stepwise multivariate logistic regression analyses. For participants who completed 3-month assessments and then were lost to follow-up, assessments at last follow-up were used for analysis. *JMP* software (SAS) was used for statistical analysis.

Results

41 participants were consented and enrolled. Out of these, 1 declined further participation in the study prior to completion of testing, and 1 had significant worsening of NIHSS prior to testing. 39 participants were able to complete at least some portions of inpatient cognitive testing and 36 participants were available for 3-month follow up. Demographically, median age was 55 years (interquartile range 46–60), 77.5% were male, and 22.5% were female. 57% of patients returned to work, 77% returned to driving, 71% had mRS of 0–1 at 3 months (Table 1).

For the 3-month follow-up in univariate analysis, younger age, white matter disease, lower NIHSS, MOCA, CDT, and normal HADS were significantly associated with return to work. MOCA and CDT were associated with return to driving. MOCA, CDT, and normal HADS were associated with mRS of 0–1 (Table 2). For multivariate analysis at 3 months, a single point increase in CDT was associated with an OR of 3.79 for return to work (95% CI 1.10–14.14) and an OR of 6.74 for return to driving (95% CI 1.22–37.23). Higher MOCA scores had an OR

Table 1. Patient and outcome characteristics.

Age mean (sd)	51.67 (9.94)
Male	77.49%
White collar occupation (vs Blue collar)	53.85%
Years of education mean (sd)	13.45 (2.13)
Prestroke mRS 1 (vs 0)	7.69%
Cortical stroke location (vs noncortical) [<i>n</i> = 37]	27.03%
Right stroke laterality (vs left/bilateral) [<i>n</i> = 37]	40.54%
Presence of white matter disease	48.72%
NIHSS mean (sd)	1.54 (0.94)
MOCA score mean (sd)	23.36 (3.77)
CDT score mean (sd) [<i>n</i> = 36]	8.44 (1.76)
TMT-A seconds mean (sd) [<i>n</i> = 36]	53.39 (25.70)
TMT-B seconds mean (sd) [<i>n</i> = 33]	111.43 (63.44)
Backwards digit span score mean (sd) [<i>n</i> = 38]	3.58 (1.72)
HADS abnormal	30.77%
Working at 3 months [<i>n</i> = 36]	57.14%
Driving at 3 months [<i>n</i> = 36]	77.14%
mRS ≤ 1 at 3 months [<i>n</i> = 36]	71.43%
Working at 6 months [<i>n</i> = 34]	66.67%
Driving at 6 months [<i>n</i> = 34]	66.67%
mRS ≤ 1 at 6 months [<i>n</i> = 34]	66.67%
Working at 12 months [<i>n</i> = 34]	72.73%
Driving at 12 months [<i>n</i> = 34]	78.79%
mRS ≤ 1 at 12 months [<i>n</i> = 34]	69.70%

Definitions: sd = standard deviation; mRS = modified Rankin Scale; NIHSS = National Institutes of Health Stroke Scale; MOCA = Montreal Cognitive Assessment; CDT = Clock Drawing Test; TMT-A = Trail Making Test – A; TMT-B = Trail Making Test – B; HADS = Hospital Anxiety and Depression Scale

of 1.56 for having an mRS of 0–1 (95% CI 1.00–2.45), and a normal HADS had an OR of 14.77 for having an mRS of 0–1 (95% CI 1.05–207.06) (Table 3).

Between 3 and 6 months, 2 participants expired (1 related to a second stroke; 1 related to an acute MI) and were not included in analysis. For 34 participants at 6-month follow-up in univariate analysis, MOCA and normal HADS were associated with return to work, driving, and mRS of 0–1. In addition, CDT and presence of white matter disease on imaging were also associated with return to driving and mRS. However, in multivariate analysis, only MOCA was associated with return to work at 6 months with an OR of 1.35 (95% CI 1.03–1.77) (Table 4).

For 12-month follow-up in univariate analysis, MOCA and CDT were associated with return to work and an mRS of 0–1, but not with return to driving. In addition, presence of white matter disease was associated with return to work, and NIHSS was associated with mRS of 0–1. In multivariate analysis, only CDT was associated with return to work at 12 months with an OR of 8.67 (95% CI 1.00–75.02) (Table 5).

Table 2. Univariate analysis of outcomes at 3 months.

	Return to Work		Return to Driving		mRS ≤ 1	
	OR (95% CI)	p-value	OR (95% CI)	p-value	OR (95% CI)	p-value
Age	0.91 (0.83–1.00)	0.05	0.91 (0.78–1.01)	0.13	0.93 (0.84–1.03)	0.16
Male	4.75 (0.78–29.02)	0.09	1.53 (0.19–9.35)	0.66	5.11 (0.89–29.29)	0.07
White Collar	3.25 (0.81–13.03)	0.10	1.92 (0.39–10.89)	0.42	3.18 (0.67–15.15)	0.15
Education	1.56 (0.98–2.49)	0.06	1.63 (0.98–3.41)	0.12	1.24 (0.82–1.89)	0.31
Prestroke mRS	0.7 (0.04–12.16)	0.81	0.26 (0.01–4.48)	0.36	0.36 (0.02–6.38)	0.49
Cortical location	3.79 (0.65–21.96)	0.14	3.11 (0.33–29.66)	0.32	1.65 (0.28–9.79)	0.58
Stroke laterality						
Left/bilateral	3.6 (0.51–33.57)	0.20	12 (0.98–172.95)	0.05	1.83 (0.22–15.33)	0.58
Right/bilateral	2.66 (0.38–24.34)	0.33	2.5 (0.35–18.04)	0.36	0.9 (0.12–6.78)	0.92
Left/right	1.35 (0.29–6.42)	0.70	5.2 (0.50–54.06)	0.17	2.04 (0.38–10.94)	0.41
White matter disease	0.22 (0.05–0.95)	0.04	0.29 (0.05–1.69)	0.17	0.18 (0.03–1.04)	0.06
NIHSS	0.43 (0.17–0.92)	0.04	0.74 (0.30–1.72)	0.48	0.46 (0.20–1.09)	0.08
MOCA	1.44 (1.14–2.00)	0.01	1.37 (1.05–1.78)	0.02	1.7 (1.18–2.45)	<0.01
CDT	2.81 (1.51–6.78)	0.01	8.26(2.39–133.25)	0.02	1.96 (1.13–3.40)	0.02
TMT-A	0.98 (0.95–1.11)	0.40	0.98 (0.94–1.11)	0.23	0.98 (0.95–1.02)	0.33
TMT- B	0.99 (0.98–1.01)	0.34	0.99 (0.97–1.01)	0.27	0.99 (0.98–1.00)	0.21
Backwards digit span	1.17 (0.77–1.83)	0.46	0.98 (0.58–1.65)	0.95	1.26 (0.78–2.04)	0.34
Normal HADS	10.28 (1.73–60.90)	0.01	1.71 (0.32–9.11)	0.53	7.88 (1.50–41.27)	0.01

Definitions: OR = odds ratio; CI = confidence interval; mRS = modified Rankin Scale; NIHSS = National Institutes of Health Stroke Scale; MOCA = Montreal Cognitive Assessment; CDT = Clock Drawing Test; TMT-A = Trail Making Test – A; TMT-B = Trail Making Test – B; HADS = Hospital Anxiety and Depression Scale

Table 3. Multivariate logistic regression of outcomes at 3 months.

	OR (95% CI)	p-value
	Return to Work	
Age	0.82 (0.65–1.04)	0.107
White matter disease	0.25 (0.01–8.54)	0.444
NIHSS	0.68 (0.09–5.20)	0.709
MOCA	1.25 (0.83–1.86)	0.285
CDT	3.79 (1.10–14.14)	0.047
Normal HADS	20.90 (0.14–3115.76)	0.234
	Return to Driving	
MOCA	1.18 (0.76–1.81)	0.454
CDT	6.74 (1.22–37.23)	< 0.001
	mRS ≤ 1	
MOCA	1.56 (1.00–2.45)	0.050
Clock drawing	1.44 (0.74–2.81)	0.287
Normal HADS	14.77 (1.05–207.06)	0.046

Definitions: OR = odds ratio; CI = confidence interval; mRS = modified Rankin Scale; NIHSS = National Institutes of Health Stroke Scale; MOCA = Montreal Cognitive Assessment; CDT = Clock Drawing Test; HADS = Hospital Anxiety and Depression Scale

Discussion

To our knowledge, there are no studies that have been reported on how bedside cognitive testing with commonly utilized assessments during acute mild ischemic stroke can predict return to work and driving. A common first question and significant goal for working-aged adults

Table 4. Multivariate logistic regression of outcomes at 6 months.

	OR (95% CI)	p-value
	Return to Work	
MOCA	1.35 (1.03–1.77)	0.032
Normal HADS	5.15 (0.83–31.99)	0.079
	Return to Driving	
White matter disease	0.16 (0.13–2.06)	0.160
MOCA	1.27 (0.85–1.92)	0.246
CDT	1.82 (0.72–4.62)	0.205
Normal HADS	3.33 (0.31–36.00)	0.322
	mRS ≤ 1	
White matter disease	0.16 (0.13–2.06)	0.160
MOCA	1.27 (0.85–1.92)	0.246
CDT	1.82 (0.72–4.62)	0.205
Normal HADS	3.33 (0.31–36.00)	0.322

Definitions: OR = odds ratio; CI = confidence interval; mRS = modified Rankin Scale; NIHSS = National Institutes of Health Stroke Scale; MOCA = Montreal Cognitive Assessment; CDT = Clock Drawing Test; HADS = Hospital Anxiety and Depression Scale

post stroke is whether they can expect to return to work and driving, as it will have a significant effect on the individual, finances, and the family quality of life. The potential for return to these important activities is difficult to determine for clinicians in the acute setting because of the limited evidence of the validity of any available test instrument to make such determination. In this prospective study, we found that CDT was most significantly

Table 5. Multivariate logistic regression of outcomes at 12 months.

	OR (95% CI) Return to Work	p-value
White matter disease	0.07 (0.00–1.71)	0.102
MOCA	0.93 (0.58–1.49)	0.753
CDT	8.67 (1.00–75.02)	0.050
	mRS ≤ 1	
NIHSS	0.03 (0.00–3.00)	0.133
MOCA	1.82 (0.59–5.59)	0.298
CDT	1.45 (0.22–9.60)	0.703

Definitions: OR = odds ratio; CI = confidence interval; mRS = modified Rankin Scale; NIHSS = National Institutes of Health Stroke Scale; MOCA = Montreal Cognitive Assessment; CDT = Clock Drawing Test

associated with early return to work and driving, and HADS was associated with early functional independence. For longer terms, MOCA was associated with return to work at 6 months, and CDT was associated with return to work at 12 months.

CDT can be administered in less than 5 min and has good potential to be feasibly used as an inpatient screening tool in ischemic stroke. Also, it relies on the patient having preservation of a broad range of visuospatial and motor skills, which may reflect the domains needed for function at work and driving. CDT has been shown previously to be predictive of development of dementia in population-based samples.²⁶ More recently, CDT has also been studied in stroke. Champod et al in 2019 evaluated CDT in acute ischemic stroke for all ages and severities and its association with long-term outcomes, including mRS and the Reintegration to Normal Living (RNL) index, which is used to determine the degree to which patients have achieved reintegration in recreation, family interaction, and community activities.¹³ It was found that better performance on CDT was associated with survival of stroke and more favorable one-year mRS and one-year RNL scores. In our study, CDT is also predictive of return to work in one year.

The MOCA has been more widely studied in stroke and has been found to be sensitive to cognitive impairment in acute TIA or mild ischemic stroke.¹¹ Dong et al also found that abnormal MOCA administered within 2 weeks of stroke correlated with moderate-severe vascular cognitive impairment as assessed by neuropsychological evaluation at 1 year.²² In another study on subarachnoid hemorrhage patients, MOCA testing done at 6 months post hemorrhage correlated with return to work at one year.²⁷ In our study, the MOCA was associated with return to work at 6 months, but no longer at one year.

TMT-A and TMT-B were included in our study due to reported previous use in assessing fitness to drive in older adults and in post neurological injury, as it tests executive function and visuomotor function.^{28–30} Similarly,

backward digit span testing assesses working memory, and has also been associated with driving cessation in mild cognitive impairment.³¹ However, these tests were not associated with any outcomes in our study. The lack of significant finding may be because the participants in our study did not uniformly have a formal driving evaluation or clearance prior to returning to drive post stroke, and the patient or family decision to start driving may not have reflected ability to safely drive without impairment. As there is evidence that poor performance on this test is associated with increased crash risk, more routine formal driving clearance in even mild stroke patients would be beneficial.^{29,32} The variable of white matter disease was studied because it has been associated with cognitive impairment and functional disability in healthy adults.³³ There has been previous data showing that nonspecific white matter changes can predict outcome after stroke,^{34,35} but this has mostly been limited to 90-day outcome. In our study, the presence of white matter disease was only associated in univariate analysis with return to driving and independent mRS at 3 months, and with return to work at 6 months and 12 months, but it was not significant in multivariate analysis in any time period. Our study represented an overall younger population with median age of 55, which may explain the lack of significant white matter burden as a major factor leading to poor outcome.

Difficulties in mood and emotional well-being are well documented post mild stroke in a significant number of survivors,^{7,36} and have also been shown to be associated with more severe cognitive impairment post-stroke with decreased likelihood of return to work.^{37,38} Anxiety and depression likely have a multi-directional relationship with stroke, with preexisting anxiety in one study correlating with an increased emergence of post-stroke anxiety,³⁹ but also the occurrence of stroke leading to potential damage in neurotransmitter pathways and causing psychosocial challenges that increase risk of psychiatric disease.⁴⁰ In our study, normal HADS score was associated in multivariate analysis with independent functioning at 3 months, but not at longer time periods, and not with return to work or driving. This is an area of further investigation, as anxiety and depression scales may be beneficial to follow long term to evaluate their fluctuation post-stroke and the effect on return to normal function. Pharmacological treatment targeting mood in early stroke may be a feasible intervention to improve early return to independent function and former activities, as shown previously with evidence regarding improvement of motor function.^{41,42}

The finding that cognitive tests familiar to clinicians and quickly completed in an inpatient setting may be able to identify those who may not return to work, driving, or functional independence is clinically significant. In addition to guiding expectations of recovery and family support, cognitive testing predictors can serve as a basis for

specifically tailored rehabilitation interventions which can potentially improve return to work, driving, and previous levels of function. Rehabilitation services may include interventions involving exercise, diet, and specialized occupational, speech, cognitive therapy. There is evidence that structured physical activity can lead to better cognitive function post-stroke from randomized control trials.⁴³ The MIND diet in stroke survivors has also been found to be associated with a slower rate of cognitive decline.⁴⁴ In addition, using a driving simulator for training post-stroke can also potentially lead to better driving fitness.⁴⁵ Finally, the use of vocational therapy after mild stroke needs further investigation, as there has been some evidence that specialized therapy interventions geared towards workplace reinvolverment can improve outcomes in return to work after stroke.⁴⁶ Further research on development of a stroke-specific vocational rehabilitation is needed.^{47,48}

There are some limitations to this study, which was designed as an exploratory study to find the cognitive testing measures most predictive of the outcome. The prospective design focusing on only mild stroke survivors within one year using strict NIHSS criteria limited the numbers enrolled, which led to lower power for analysis. This can be broadened in future studies and interventional trials by enrolling larger numbers and increasing the NIHSS criteria needed for enrollment.

Conclusions

Bedside cognitive testing after acute ischemic stroke is shown to predict functional outcomes, including return to work, driving, and independence with daily activities at early and long term follow up intervals. These relatively easily implemented tools can help determine prognosis and identify patients who may benefit from further interventions including specialized rehabilitation programs. Further studies with larger numbers of mild stroke patients are needed to validate these findings, and to identify combinations of therapies that may meet these functional goals.

Declaration of Competing Interest

None

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